

Задача Коши для волн. ур-я (сущий 2-х пространств. переменных)

$$u(x_1, x_2, t) \quad \bar{x} = (x_1, x_2)$$

$$(1) u_{tt} - \Delta u = F(x, t), \quad x \in \mathbb{R}^2, \quad t > 0$$

$$(2) u|_{t=0} = \varphi(x), \quad x \in \mathbb{R}^2$$

$$(3) u_t|_{t=0} = \psi(x), \quad x \in \mathbb{R}^2$$

оп-я $u(x, t)$ - решение (1), (2), (3):

$$1) u(x, t) \in C^2(x \in \mathbb{R}^2, t \geq 0)$$

$$(x_1, x_2, \underline{x_3}) \in \mathbb{R}^3$$

$$2) \text{ уга. (1), (2), (3)}$$

$$I \quad u(x, t) - \text{реш.} \uparrow$$

$$\Downarrow \quad \varphi(x) \in C^2(\mathbb{R}^2), \quad \psi(x) \in C^1(\mathbb{R}^2), \quad F(x, t) \in C(x \in \mathbb{R}^2, t > 0)$$

$$u(x_1, x_2, t) \longleftrightarrow u(x_1, x_2, \underline{x_3}, t)$$

$$\varphi(x_1, x_2) \longleftrightarrow \varphi(x_1, x_2, \underline{x_3})$$

$$\psi(x_1, x_2) \longleftrightarrow \psi(x_1, x_2, \underline{x_3})$$

$$u(x_1, x_2, x_3, t)$$

$$u_{tt} - \Delta u = F(x, t), \quad x = (x_1, x_2, x_3) \in \mathbb{R}^3, \quad t > 0$$

$$u|_{t=0} = \varphi(x_1, x_2, x_3)$$

$$u_t|_{t=0} = \psi(x_1, x_2, x_3)$$

$$F(x, t) \equiv 0$$

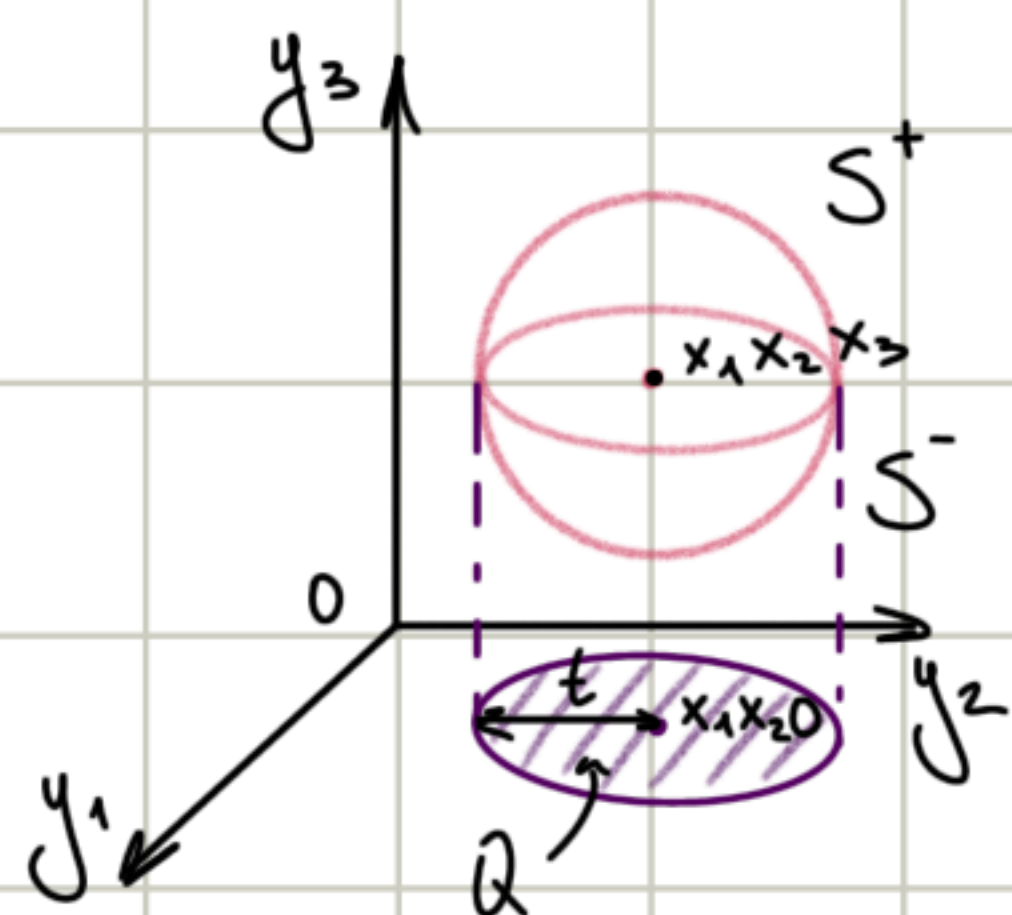
$$\varphi(x_1, x_2) \in C^3(\mathbb{R}^2) \Rightarrow \varphi(x_1, x_2, x_3) \in C^3(\mathbb{R}^3)$$

$$\psi(x_1, x_2) \in C^2(\mathbb{R}^2) \Rightarrow \psi(x_1, x_2, x_3) \in C^2(\mathbb{R}^3)$$

$$u(x_1, x_2, x_3, t) = \frac{1}{4\pi t} \iint_{|y-x|=t} \varphi(y_1, y_2, \underline{y_3}) dS_y + \frac{\partial}{\partial t} (A_\varphi(x, t))$$

$$A_\varphi(x_1, x_2, x_3, t) = \frac{1}{4\pi t} \iint_{|y-x|=t} \varphi(y_1, y_2, \underline{y_3}) dS_y \quad (=)$$

$$\iint_{S^+} = \iint_Q \varphi dS = |\bar{r}'_{y_1}, \bar{r}'_{y_2}| dy_1 dy_2$$



$$(y_1 - x_1)^2 + (y_2 - x_2)^2 + (y_3 - x_3)^2 = t^2$$

$$y_3 = x_3 \pm \sqrt{t^2 - (y_1 - x_1)^2 - (y_2 - x_2)^2}$$

$$S^+: \begin{cases} y_1 = y_1 \\ y_2 = y_2 \\ y_3 = x_3 + \sqrt{\dots} \end{cases}$$

$$(y_1, y_2) \in Q$$

$$S^-: \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ x_3 - \sqrt{\dots} \end{pmatrix} = \bar{r}(y_1, y_2)$$

$$(y_1, y_2) \in Q$$

$$= \frac{2}{4\pi t} \iint_Q \psi(y_1, y_2, y_3) / [L \bar{r}'_{y_1}, \bar{r}'_{y_2}] dy_1 dy_2 =$$

$$\bar{r}'_{y_1} = \begin{pmatrix} 1 \\ 0 \\ \frac{1}{2} \frac{-2(y_1-x_1)}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} \end{pmatrix} \qquad \bar{r}'_{y_2} = \begin{pmatrix} 0 \\ 1 \\ \frac{-(y_2-x_2)}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} \end{pmatrix}$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -\frac{y_1-x_1}{\sqrt{\dots}} \\ 0 & 1 & -\frac{y_2-x_2}{\sqrt{\dots}} \end{vmatrix} = \frac{y_1-x_1}{\sqrt{\dots}} \vec{i} + \frac{y_2-x_2}{\sqrt{\dots}} \vec{j} + 1 \cdot \vec{k}$$

$$|L \bar{r}'_{y_1}, \bar{r}'_{y_2}|^2 = \frac{(y_1-x_1)^2}{t^2-(y_1-x_1)^2-(y_2-x_2)^2} + \frac{(y_2-x_2)^2}{t^2-(y_1-x_1)^2-(y_2-x_2)^2} + 1 = \frac{t^2}{t^2-(y_1-x_1)^2-(y_2-x_2)^2}$$

$$\Rightarrow dS_{\pm} = \frac{t}{\sqrt{\dots}} dy_1 dy_2$$

$$= \frac{2}{4\pi t} \iint_Q \frac{\psi(y_1, y_2) t}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} dy_1 dy_2 = \frac{1}{2\pi} \iint_Q \frac{\psi(y_1, y_2)}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} dy_1 dy_2 =$$

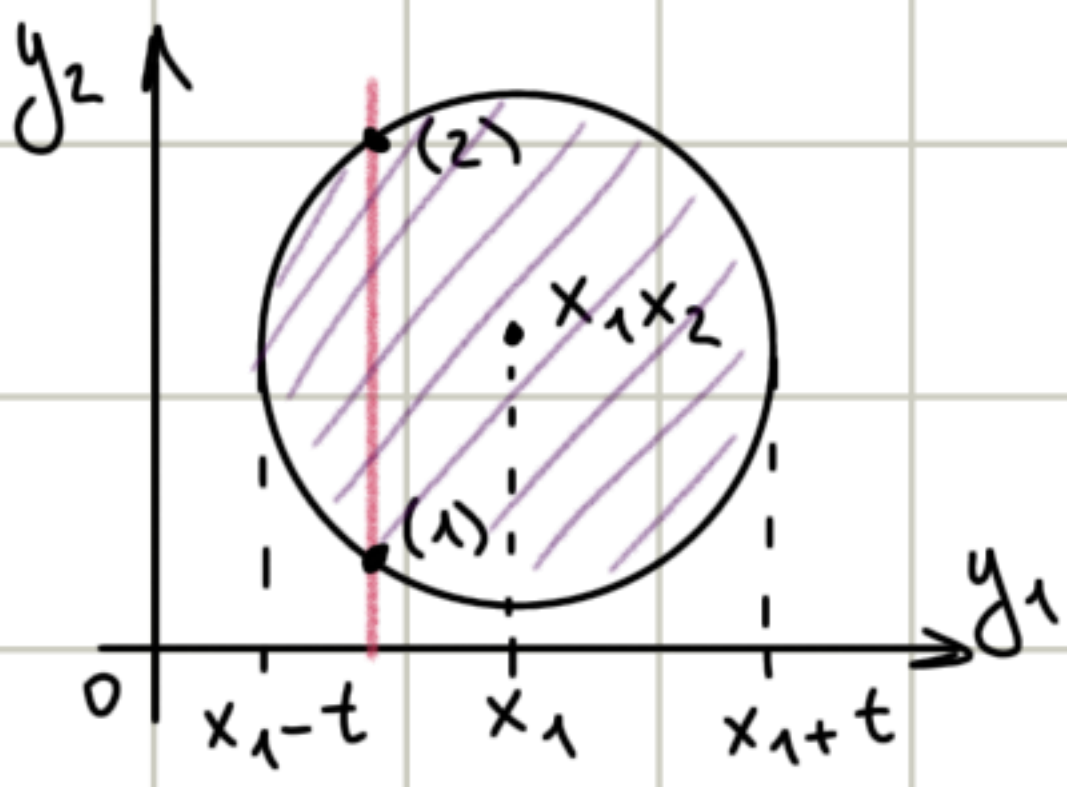
$$u(x_1, x_2, t) = \frac{1}{2\pi} \iint_Q \frac{\psi(y_1, y_2) dy_1 dy_2}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} + \frac{\partial}{\partial t} \left[\frac{1}{2\pi} \iint_Q \frac{\varphi(y_1, y_2) dy_1 dy_2}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} \right] \quad \text{ф-ла Гюассона}$$

$$u_{tt} - u_{x_1 x_1} = 0$$

$$u_{tt} - \Delta u = 0$$

$$u(x_1, t)$$

$$u(x_1, x_2, x_3, t)$$



$$x_1 - t \leq y_1 \leq x_1 + t$$

$$x_2 - \sqrt{t^2 - (y_1 - x_1)^2} \leq y_2 \leq \sqrt{t^2 - (y_1 - x_1)^2}$$

$$= \frac{1}{2\pi} \int_{x_1-t}^{x_1+t} dy_1 \left[\int_{x_2-\sqrt{t^2-(y_1-x_1)^2}}^{x_2+\sqrt{t^2-(y_1-x_1)^2}} \frac{\psi(y_1, y_2)}{\sqrt{t^2-(y_1-x_1)^2-(y_2-x_2)^2}} dy_2 \right] =$$

$$\int_{x_2-\sqrt{t^2-(y_1-x_1)^2}}^{x_2+\sqrt{t^2-(y_1-x_1)^2}} \frac{1}{\sqrt{a^2-(y_2-x_2)^2}} d(y_2-x_2) = \arcsin\left(\frac{y_2-x_2}{\sqrt{t^2-(y_1-x_1)^2}}\right) \Big|_{y_2=x_2-\sqrt{t^2-(y_1-x_1)^2}}^{y_2=x_2+\sqrt{t^2-(y_1-x_1)^2}} =$$

$$= \arcsin 1 - \arcsin(-1) = 2\arcsin 1 = \pi$$

$$= \frac{1}{2} \int_{x_1-t}^{x_1+t} \psi(y_1) dy_1$$

$$u(x_1, t) = \frac{1}{2} \int_{x_1-t}^{x_1+t} \psi(y_1) dy_1 + \frac{\partial}{\partial t} \left[\frac{1}{2} \int_{x_1-t}^{x_1+t} \varphi(y_1) dy_1 \right] = \frac{1}{2} \int_{x_1-t}^{x_1+t} \psi(y_1) dy_1 + \frac{1}{2} \left\{ \varphi(x_1+t) \cdot 1 - \varphi(x_1-t) \cdot (-1) \right\} \quad \text{ф-ла Даламбера}$$

$$u(x_1, x_2, x_3, t) \quad a > 0$$

$$u_{tt} - a^2 \Delta u = 0, \quad x = (x_1, x_2, x_3) \in \mathbb{R}^3, \quad t > 0$$

$$u|_{t=0} = \varphi(x) \in C^3(\mathbb{R}^3)$$

$$u_t|_{t=0} = \psi(x) \in C^2(\mathbb{R}^3)$$

$$\frac{1}{a^2} u_{tt} - \Delta = 0$$

$$\begin{cases} x_1 = x_1 \\ x_2 = x_2 \\ x_3 = x_3 \\ r = at \end{cases}$$

$$\text{D/3: } u(x, t) = \frac{1}{4\pi a^2 t} \iint_{|y-x|=at} \varphi(y) dS_y + \frac{\partial}{\partial t} [\dots]$$

принцип Адамара

$$u(x, t)$$

$$\begin{cases} u_{tt} + u_{xx} = 0, & x \in \mathbb{R}^1, t > 0 \\ u|_{t=0} = \varphi(x), & x \in \mathbb{R}^1 \\ u_t|_{t=0} = \psi(x), & x \in \mathbb{R}^1 \end{cases}$$

$$u|_{t=0} = 0$$

$$\begin{cases} u_{tt} + u_{xx} = 0, & x \in \mathbb{R}^1, t > 0 \\ u|_{t=0} = 0 =: \varphi_0(x) \\ u_t|_{t=0} = 0 =: \psi_0(x) \end{cases}$$

$$u_0(x, t) \equiv 0$$

$$u|_{t=0} = k$$

$$\begin{cases} u_{tt} + u_{xx} = 0, & x \in \mathbb{R}^1, t > 0 \\ u|_{t=0} = 0 =: \varphi_k(x) \\ u_t|_{t=0} = \frac{1}{k} \cos kx =: \psi_k(x) \end{cases}$$

$$|\psi_k(x) - \psi_0(x)| = \frac{1}{k} |\cos kx| \leq \frac{1}{k}$$

$$\psi_k(x) \xrightarrow[k \rightarrow \infty]{\mathbb{R}^1} \psi_0(x)$$

$$u_k(x, t) = f(t) \cos kx = (A \cosh kt + B \sinh kt) \cos kx \quad \forall A, \forall B$$

$$f''(t) \cos kx + f(t) (-k^2) \cos kx = 0$$

$$f''(t) - k^2 f(t) = 0$$

$$\begin{aligned} \lambda^2 - k^2 &= 0 & e^{kt}, e^{-kt} \\ \pm k & & \cosh kt, \sinh kt \end{aligned}$$

$$t=0: A \cos kx = 0 \Rightarrow A=0$$

$$Bk \cos kx = \frac{1}{2} \cos kx$$

$$B = \frac{1}{k^2}$$

$$u_k(x, t) = \frac{1}{k^2} \sinh kt \cos kx$$

$$|u_k(x, t) - u_0(x, t)| = \left| \frac{1}{2k^2} (e^{kt} - e^{-kt}) \cos kx \right| \geq \varepsilon_0$$

$$\hat{x}, \hat{t}$$

$$\hat{x} = 0, |u_k(\hat{x}, t)| = \left| \frac{1}{2k^2} (e^{kt} - e^{-kt}) \right|$$

$$\hat{t} = \frac{1}{k}, u_k(\hat{x}, \hat{t}) = \frac{1}{2k^2} (e^1 - e^{-1}) = \varepsilon_0$$

$$(0, \frac{1}{k})$$

